



Mirpur University of Science & Technology(MUST)

Department of software engineering

Lab task#3

Subject:

SOE

Submitted To:

Engr.Asim Javid

Submitted By:

Name	Roll No	Section
Hira Shahid	FA23-BSE-029	A

EXPERIMENT NO: 03

Topic: Test planning and Management using jira

Test plan Identifier:

Test Plan ID: LMS-TP-001 v1.0

Version: 1.0

Date: May 5, 2026

Project: Library Management System (M1 & M2)

Prepared By: Hira Shahid_SQE Lab Student

Reviewed By: Engr.Asim Javid

Status: Draft — pending review

Introduction:

This document describes the Test Plan for the Authentication and Book Search modules of the Library Management System. The goal is to ensure that the security features and search functionalities meet the specified quality standards.

Test items:

Features to be tested:

Feature	Requirements Covered	Test Priority
User login with valid credentials (email + password)	REQ-01	High
Rejection of invalid login attempts (wrong password, nonexistent email)	REQ-01	High
Account lockout after exactly 3 consecutive failed attempts	REQ-02	High

Item	Version / Build	Description
LMS Frontend (React.js)	v1.0 (build #42)	Browser-rendered React application served from https://lms-staging.uni.edu
LMS Backend (Node.js/Express)	v1.0	REST API server; endpoints documented in LMSAPI-SPEC-v1.0.pdf
LMS Database (PostgreSQL)	v14.2	Relational database containing book, loan, student, and notification data
Authentication Service	v1.0.0	JWT-based authentication module handling login, session, and password reset
Test Data Set	v1.0 (LMS-TESTDATA-001)	Pre-populated database with 200 books, 50 student accounts, 5 staff accounts

Feature	Requirements Covered	Test Priority
Account logout message display and admin unlock process	REQ-02	Medium
Password reset email request and delivery within 2 minutes	REQ-03	Medium
Password reset link validity and expiry (24-hour link life)	REQ-03	Medium
Session automatic expiry after 30 minutes of inactivity	REQ-04	High
Session timer reset upon any user interaction before 30 minutes	REQ-04	Medium
Book search by title keyword (partial and full match)	REQ-05	High
Book search by author name (partial and full match)	REQ-05	High
Book search by ISBN (exact match only)	REQ-05	High
Filter search results by genre category	REQ-06	Medium
Filter search results by availability status (Available / On Loan)	REQ-06	Medium
Search response time under 2 seconds with standard test data set	REQ-07	High
Pagination of results exceeding 20 items	REQ-08	Low

Features not to be Tested:

Feature Not Tested	Justification
M3: Loan Management (REQ-09 to REQ-12)	Out of scope for Week 3 test cycle. Covered in Test Plan LMS-TP-002 (Week 4 lab).
M4: Notifications (REQ-13 to REQ-16)	Out of scope for Week 3. Covered in Test Plan LMS-TP-003. SMS gateway not available in test environment.
Load/stress testing (>100 concurrent users)	No load testing tool configured in this lab environment. Planned for Week 10.
Browser compatibility testing (Safari, Firefox, Edge)	Chrome v120+ is the designated test browser. Cross-browser testing planned for Week 8.
Mobile/responsive design testing	System is desktop-only in v1.0. Mobile version planned for v2.0.
Database backup and recovery procedures	Infrastructure testing — out of scope for application testing. Covered by DevOps team.
API security penetration testing	Requires specialized tooling (e.g., Burp Suite). Planned for Week 12 Security Testing lab.

Approach (testing strategy):

Aspect	Approach for LMS Week 3 Test Cycle
Testing Level	System testing — the entire integrated application (not unit or integration tests)
Testing Type	Primarily functional black-box testing; performance testing for REQ-07 and REQ16
Execution Method	Manual test execution — automated execution planned for Week 9 onward
Black-Box Techniques	Equivalence Partitioning (login input validation); Boundary Value Analysis (session timeout, response times)
Test Priority Order	High priority test cases executed first (REQ-01, REQ-02, REQ-04, REQ-07); then Medium, then Low
Defect Management	Defects logged in JIRA using the standard defect report template; severity assigned per defect classification guide
Re-testing	All defects fixed by development team will be re-tested within 24 hours of fix deployment
Regression Testing	After each defect fix: execute smoke test (TC-001 to TC-005) to verify no regression in core functionality

Aspect	Approach for LMS Week 3 Test Cycle
Test Data Management	Pre-loaded test database (LMS-TESTDATA-001); test data restored between test sessions using database snapshot

Item pass/fail criteria:

Level	Criterion	Pass Condition	Fail Condition
Test Case	Individual test case result	Actual result exactly matches the Expected Result documented in the test case	Any deviation from the Expected Result, including partial matches or incorrect data
Test Cycle (Exit)	Completion conditions for this test plan	100% of High priority test cases executed; 100% of Medium priority executed; 0 open Critical defects; 0 open High defects; test report reviewed and signed off	Any open Critical defect; more than 10% of High priority test cases not executed; test report not reviewed within 5 business days
Defect Severity — Critical	System crash or complete feature failure	N/A — critical defects immediately halt testing	Login module cannot be accessed; system crashes on startup; data corruption observed
Defect Severity — High	Major functional defect with no workaround	Feature works but with non-critical deviation	Account lockout not working; session does not expire; search returns no results
Defect Severity — Medium	Functional defect with available workaround	Minor deviation with documented workaround	Error message wording incorrect; pagination controls misaligned
Defect Severity — Low	Cosmetic or minor UI issue	Cosmetic defect does not affect function	Spelling error in UI text; minor layout issue on non-critical page

Suspension criteria&resumption requirements:

Suspension Criterion	Threshold	Action Required
Critical defect blocking all test cases	1 or more open Critical defects in the login module (REQ-01)	Halt all testing. Notify development lead immediately. Resume only after fix is deployed and verified
High defect rate	More than 20% of test cases in current sprint are failing with High severity defects	Pause testing. Hold defect triage meeting within 24 hours. Resume after defect resolution plan agreed
Test environment unavailable	LMS staging server unreachable for more than 2 hours during a test session	Suspend test session. Contact DevOps. Resume when environment is verified stable
Test data corruption	Test database state deviates from LMS-TESTDATA-001 baseline	Halt testing. Restore database from snapshot. Re-verify test environment before resuming
Build quality too low	More than 30% of smoke test cases (TC-001 to TC-005) fail on a new build	Reject the build. Do not proceed with further testing on this build. Request a new build from development

Test deliverables:

Deliverable	Description	Format	Due Date
Requirement Traceability Matrix	Complete RTM mapping all 16 LMS requirements to test cases with coverage status	Excel (xlsx)	End of Week 3 lab
IEEE 829 Test Plan (LMS-TP-001)	This document — test plan for M1 and M2 system testing	PDF (pdf)	End of Week 3 lab
Test Case Specification	Full test case documents for TC-001 to TC-026 (all M1 and M2 test cases)	JIRA + PDF export	End of Week 4 lab
Test Data Specification	Documentation of test data sets and setup procedures for LMS-TESTDATA-001	Excel (xlsx)	Week 4
Test Execution Report	Record of all test case execution results, pass/fail status, and defects found	PDF (pdf)	Week 5
Defect Report	JIRA defect log with all defects found during M1/M2 system testing	JIRA + CSV export	Week 5
Test Closure Report	Summary of testing completed, quality metrics, lessons learned	PDF (pdf)	Week 6

Environmental needs:

Category	Requirement	Specification
Hardware	Test machine	Laptop or desktop; minimum: Intel i5, 8GB RAM, 256GB SSD
Operating System	Test OS	Windows 10/11 or macOS 12+ or Ubuntu 22.04 LTS
Browser	Test browser	Google Chrome v120 or higher (primary); Firefox v120 (secondary for smoke only)
Network	Connectivity	Stable internet connection; minimum 10 Mbps download for accessing LMS staging server
Test Environment	Staging server	LMS staging environment: https://lms-staging.uni.edu — isolated from production; accessible only on university network or VPN
Database	Test database	PostgreSQL v14.2; pre-loaded with LMS-TESTDATA-001; snapshot backup available for restore
Email Testing	Email test tool	Gmail test account (lms.test.student@gmail.com) for password reset and notification testing
Test Management	Tool	JIRA Software — workspace: [yourname].atlassian.net; project: SQELab-[YourName]
Timer / Stopwatch	Timing tool	Mobile phone stopwatch or browser timer for performance measurements (REQ-07, REQ-16)

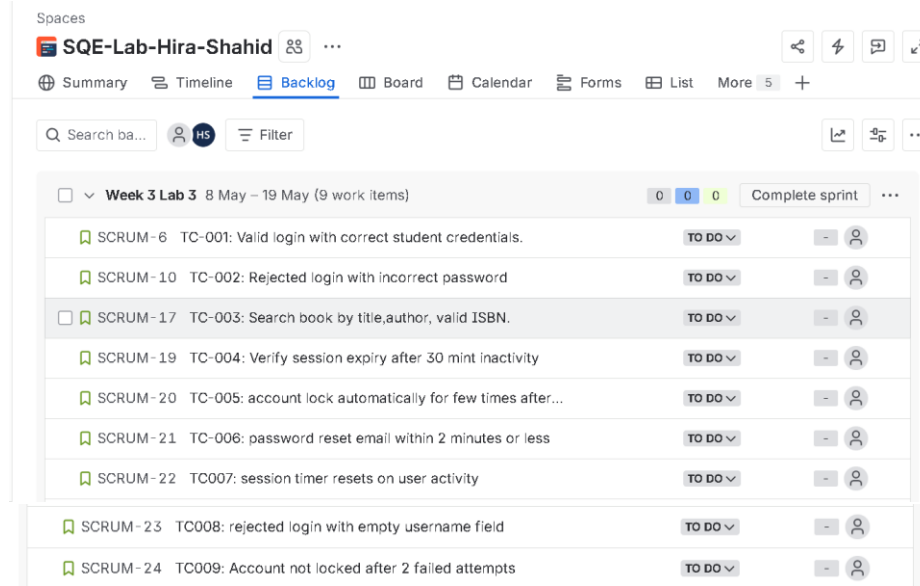
Responsibilities:

Role	Person / Team	Responsibilities
Test Manager	Hira Tanveer	Write and maintain test plan; manage RTM; report testing progress; sign off test report
Test Designer	Hira Tanveer	Design test cases for M1 and M2; create test data; review test cases
Test Executor	Hira Tanveer	Execute test cases per schedule; log defects; perform retesting after fixes
Development Team	LMS Dev Team	Provide stable builds; fix defects within agreed SLA; deploy to staging environment
Environment Administrator	DevOps / Lab Instructor	Maintain LMS staging environment; restore test data snapshots; manage server availability
Test Reviewer	Lab Instructor	Review and approve test plan and RTM; review test closure report
Business Stakeholder	University Library Management	Provide clarification on requirements; participate in UAT (Week 6)

Schedule:

Week	Phase	Activity	Deliverable
Week 3 (Lab 3)	Requirements & Planning	Configure JIRA; build RTM; write Test Plan LMS-TP-001	RTM + Test Plan
Week 4 (Lab 4)	Test Case Design	Write TC-001 to TC-026 using black-box techniques (EP + BVA)	Test Case Specification
Week 5 (Lab 5)	Test Execution	Execute M1 test cases (TC-001 to TC013); log defects	Test Execution Report (M1)
Week 6 (Lab 6)	Test Execution	Execute M2 test cases (TC-014 to TC026); log defects	Test Execution Report (M2)
Week 7 (Lab 7)	Defect Management	Retest fixed defects; update defect log; defect metrics	Defect Closure Report
Week 8 (Lab 8)	Closure	Write test closure report; lessons learned; archive artefacts	Test Closure Report (M1 + M2)

Screenshots:



Link:

<https://hirashahid756-1778128366578.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog>